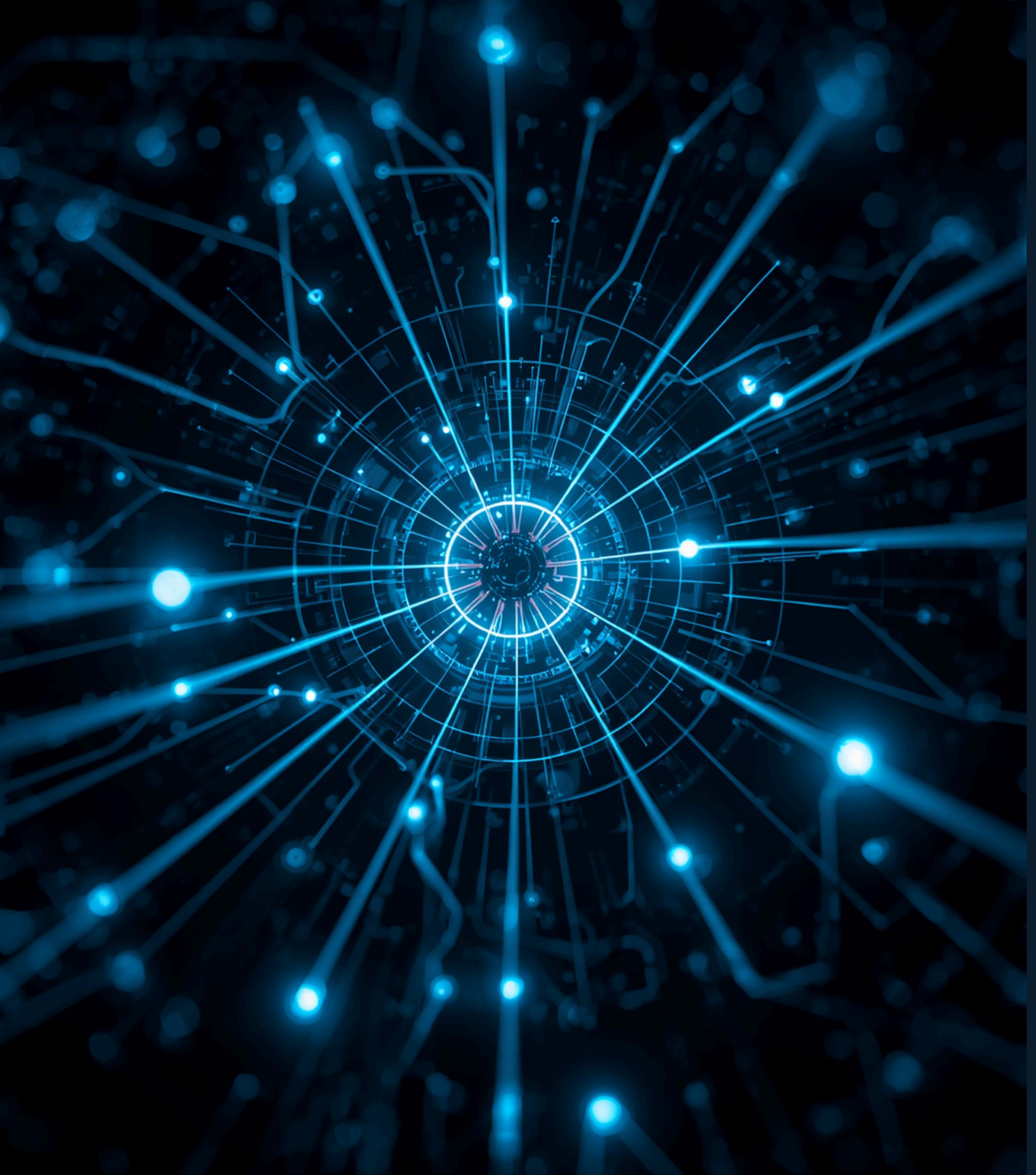


INPT

Adversarially Robust

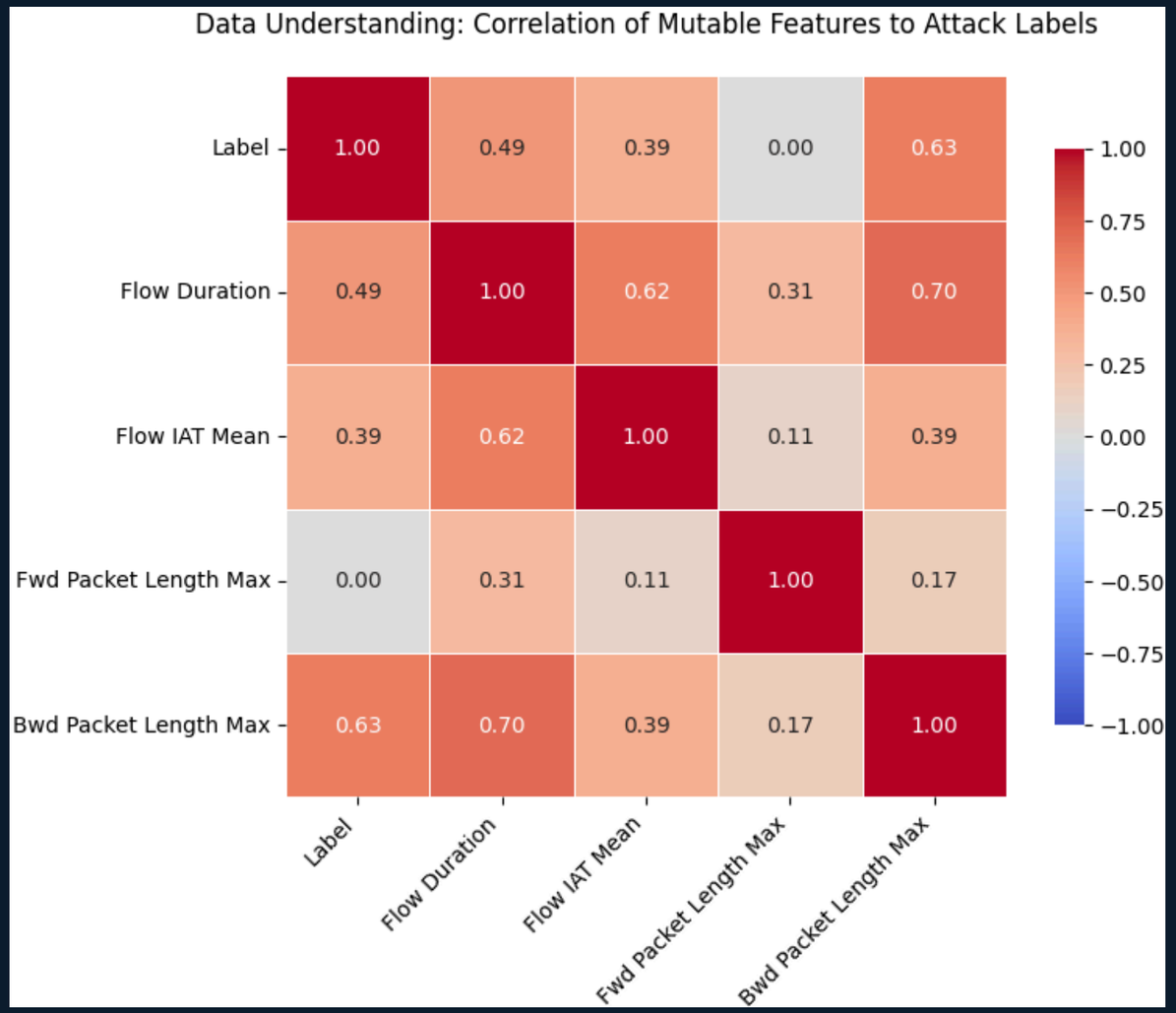
Bridging Lab Accuracy & Adversaries

PRESENTED BY

Zakaria Bigane

Marwan moujane

Forcing Behavioral Learning

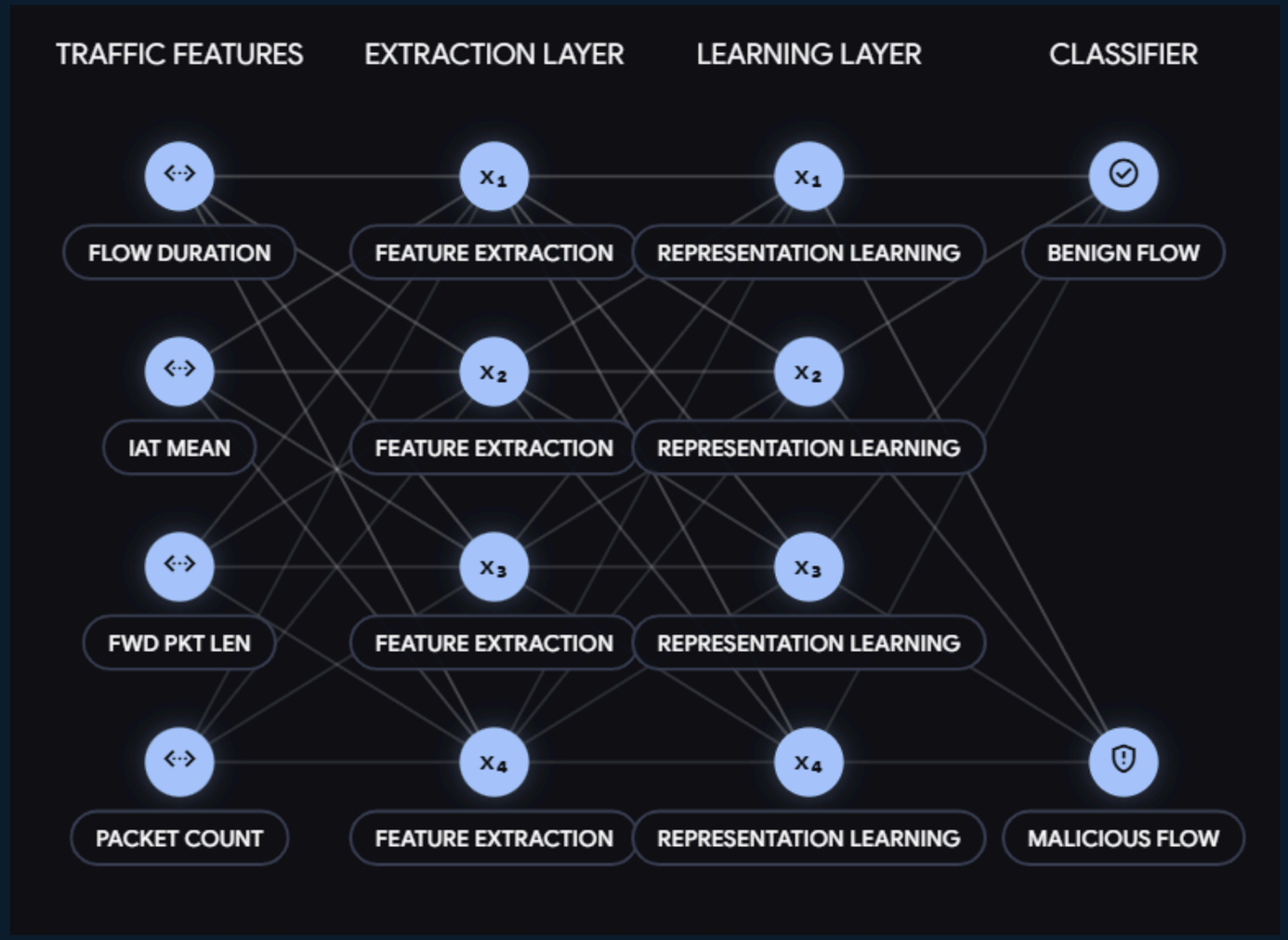


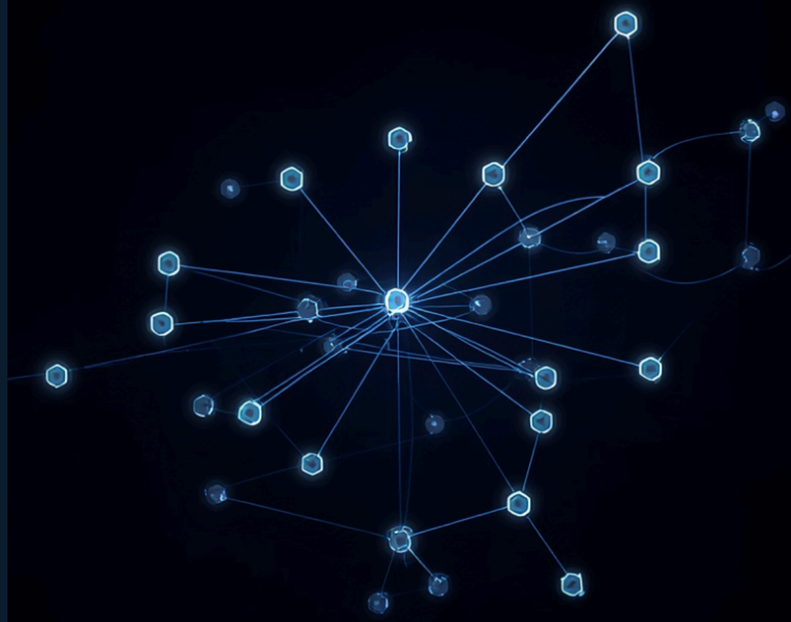
Utilizing the **CIC-IDS2017** Wednesday-workingHours subset, we've dropped host indicators to guide the model in learning behavioral patterns instead of specific instance memorization, enhancing generalization effectively.



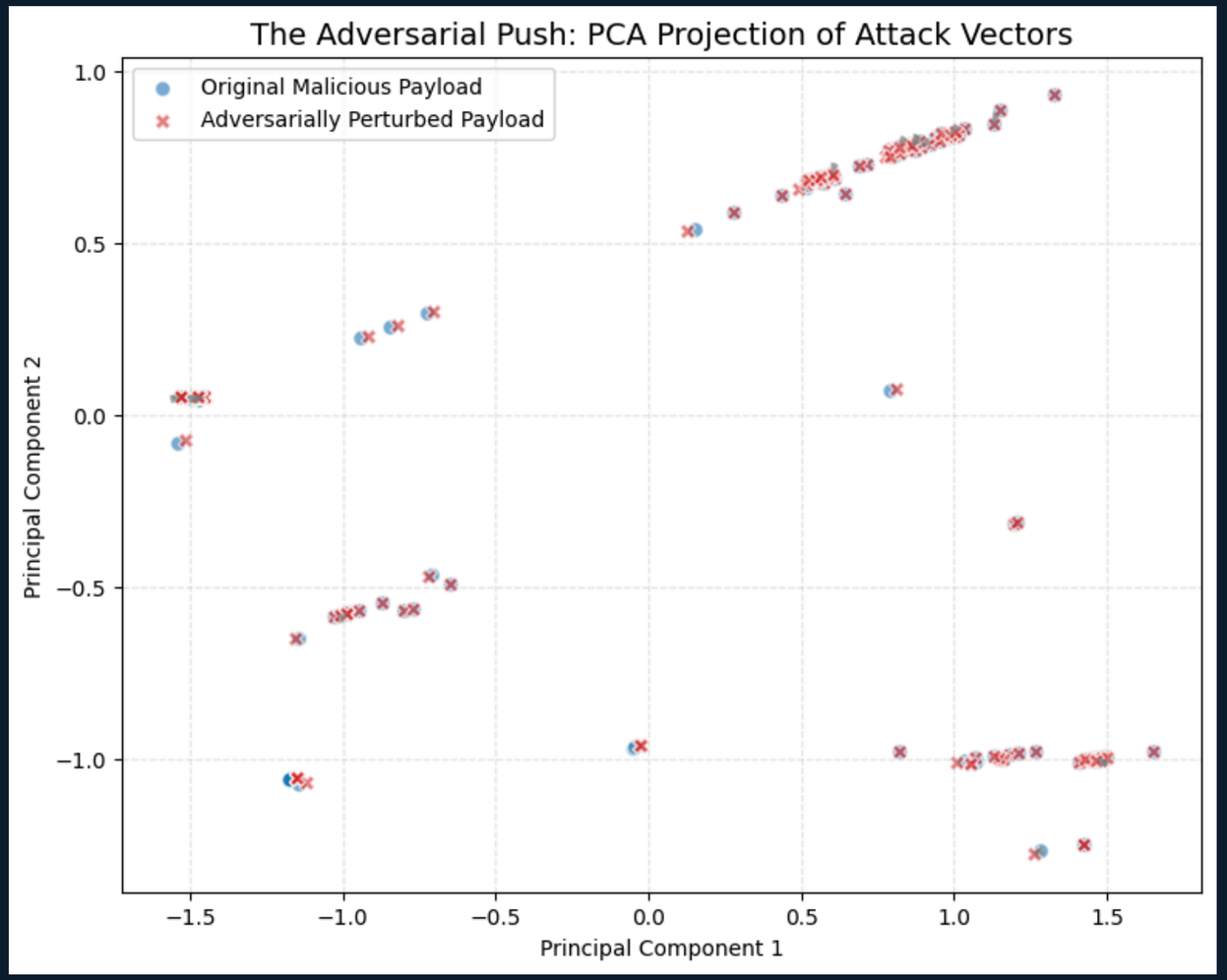
Lightweight MLP

A lightweight Multi-Layer Perceptron (MLP) architecture is designed to remain within a strict 2-hour training limit, facilitating reproducibility and practical deployment in real-world applications.





Threat Modeling



IMMUTABLE FEATURES

The 74 immutable features, such as protocol flags and destination ports, are crucial for ensuring network integrity and maintaining consistent detection capabilities against adversarial manipulations.

MUTABLE FEATURES

The 4 mutable features, including packet length max and timing characteristics, are vulnerable to adversarial manipulation, which can significantly impact the performance and reliability of intrusion detection systems.

Attack Vector

Employing a **Constrained Fast Gradient Sign Method** (FGSM) attack with epsilon 0.05, perturbations translate into network artifacts like packet padding (NOP sleds) and introduce artificial delays in data flow.



Fragility Under Adversarial Pressure

The baseline model achieves an impressive **99% F1-score on clean traffic**, yet it faces a significant spike in false negatives when exposed to adversarial attacks, highlighting its vulnerabilities.

```
================================================================
 EXPERIMENTAL REPORT: [Adversarial Training: False] | Epsilon: 0.05
================================================================
>>> UNPERTURBED DATASET METRICS:
              precision    recall  f1-score   support

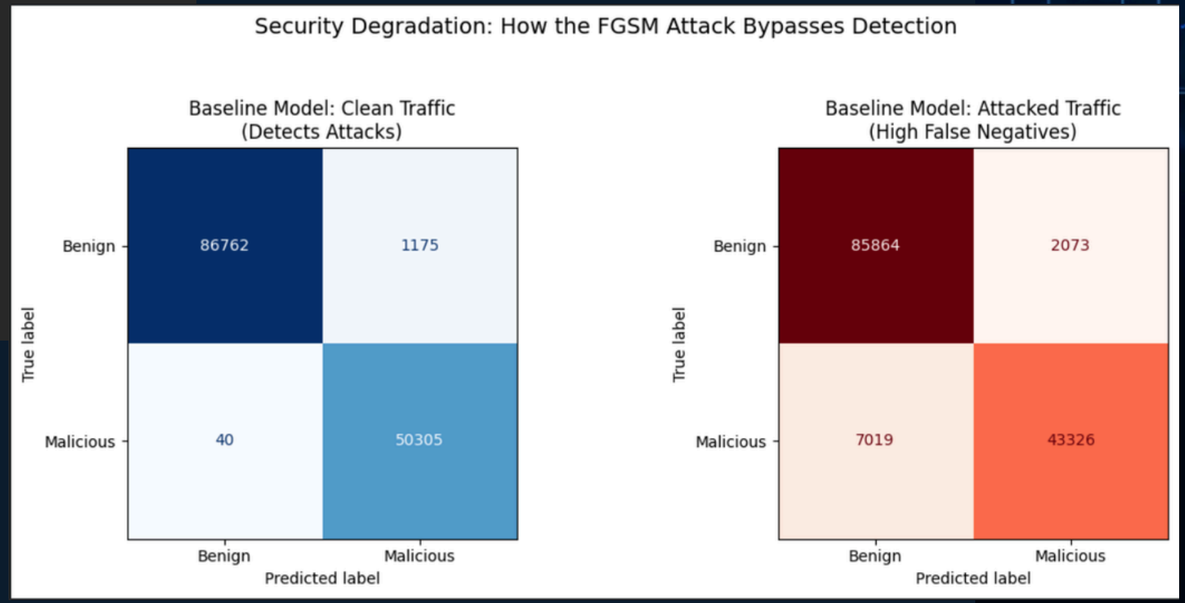
         0.0       1.00      0.99      0.99     87937
         1.0       0.98      1.00      0.99     50345

    accuracy                           0.99    138282
   macro avg       0.99      0.99      0.99    138282
weighted avg       0.99      0.99      0.99    138282

>>> ADVERSARIALLY MANIPULATED METRICS:
              precision    recall  f1-score   support

         0.0       0.92      0.98      0.95     87937
         1.0       0.95      0.86      0.91     50345

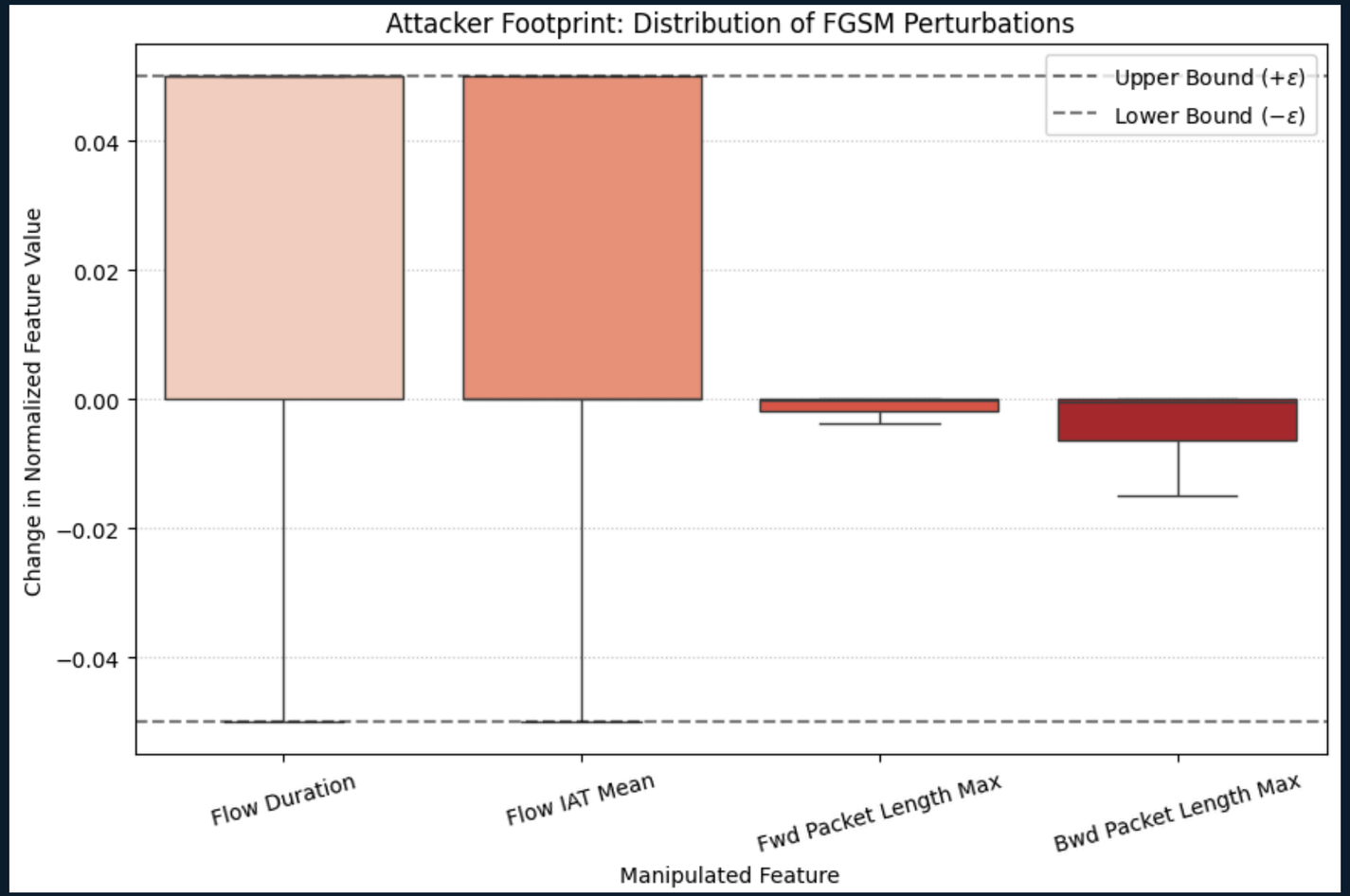
    accuracy                           0.93    138282
   macro avg       0.94      0.92      0.93    138282
weighted avg       0.94      0.93      0.93    138282
```





Defense Strategy

Implemented an **adversarial training pipeline** with dynamically generated batches to realistically expose the model to diverse attack scenarios during the learning process, enhancing overall robustness.





Hardened Model Performance

The hardened model achieves a **remarkable 99.1% F1-score** under adversarial attack, effectively maintaining baseline accuracy on normal traffic, showcasing enhanced resilience against threats.

```
============================================================
 EXPERIMENTAL REPORT: [Adversarial Training: True] | Epsilon: 0.05
============================================================
>>> UNPERTURBED DATASET METRICS:
              precision    recall  f1-score   support

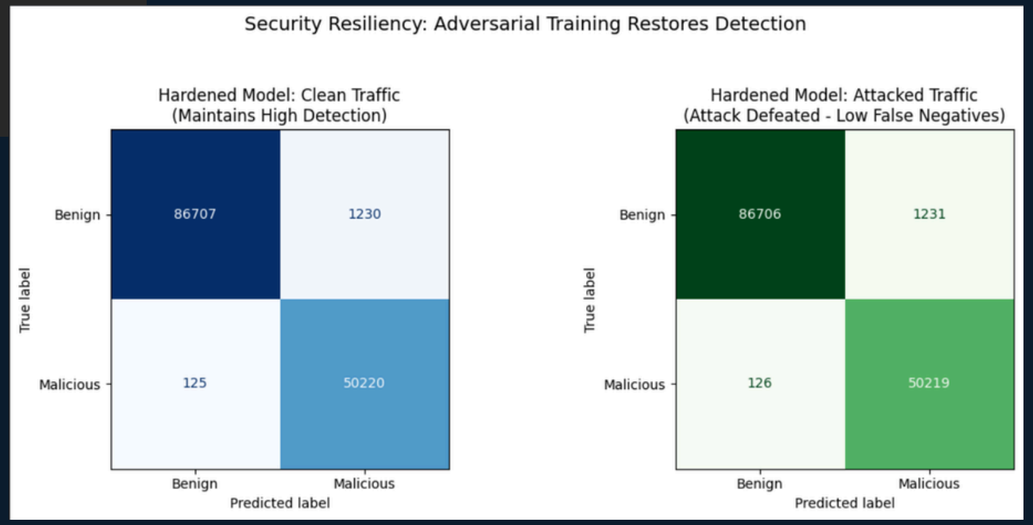
         0.0       1.00      0.99      0.99     87937
         1.0       0.98      1.00      0.99     50345

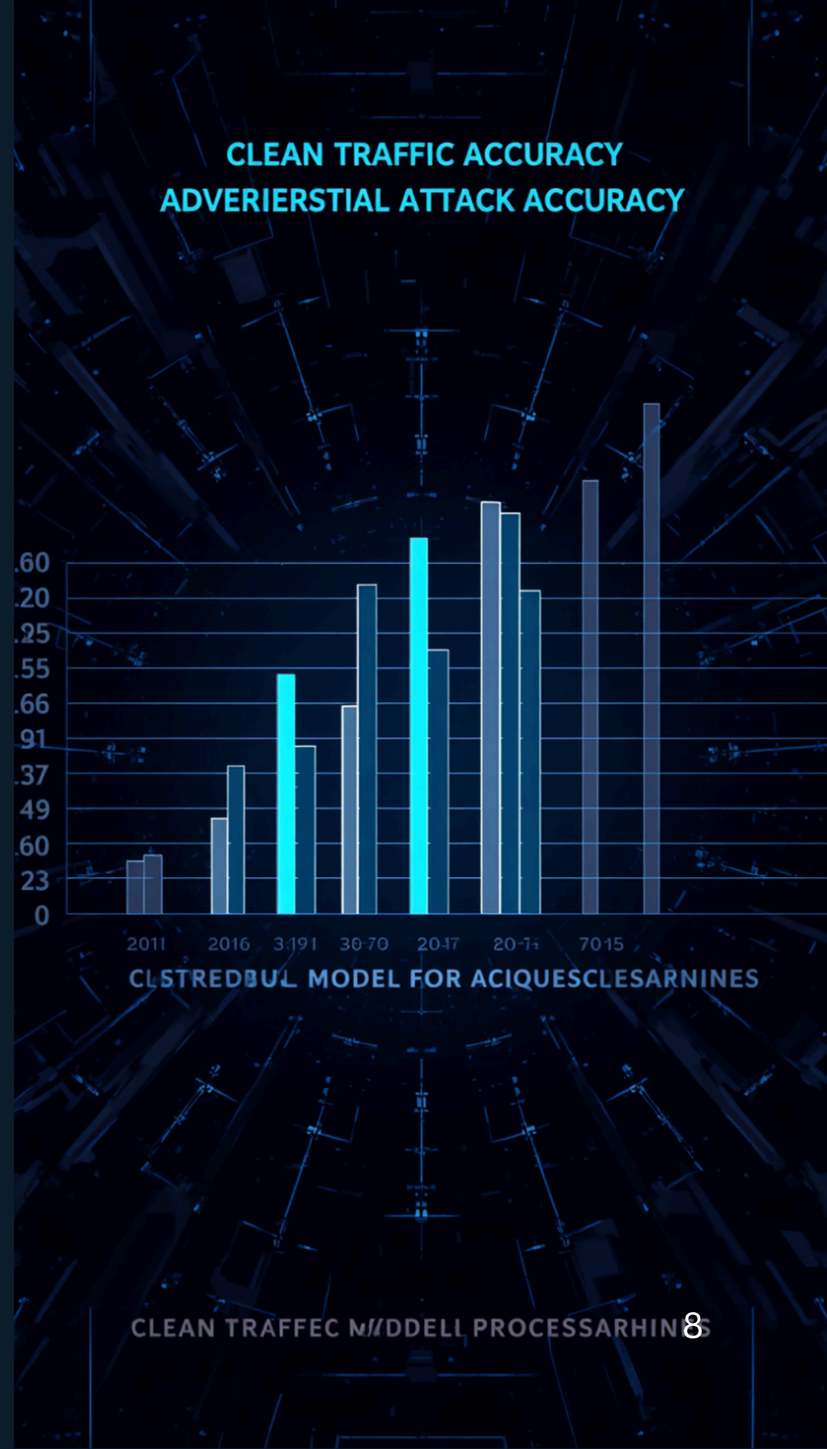
    accuracy                           0.99    138282
   macro avg       0.99      0.99      0.99    138282
weighted avg       0.99      0.99      0.99    138282

>>> ADVERSARIALLY MANIPULATED METRICS:
              precision    recall  f1-score   support

         0.0       1.00      0.99      0.99     87937
         1.0       0.98      1.00      0.99     50345

    accuracy                           0.99    138282
   macro avg       0.99      0.99      0.99    138282
weighted avg       0.99      0.99      0.99    138282
```





Conclusion & Deployment

